

Fraction Products

Every graded set, every week — 5 weeks, 25 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Fraction of a Number

Divide by the bottom, multiply by the top.

Warm-Up 8 ITEMS • RECALL

1 $\frac{1}{2}$ of 102 $\frac{1}{3}$ of 93 $\frac{1}{4}$ of 124 $\frac{1}{5}$ of 205 $\frac{1}{2}$ of 306 $\frac{1}{6}$ of 187 $\frac{1}{3}$ of 128 $\frac{1}{4}$ of 16

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

9. $\frac{2}{3}$ of 1210. $\frac{3}{4}$ of 2011. $\frac{2}{5}$ of 3512. $\frac{5}{6}$ of 2413. $\frac{3}{8}$ of 4014. $\frac{2}{3}$ of 15

Fraction × Fraction

Multiply across. Type answers like 1/6.

Warm-Up 8 ITEMS • RECALL

1 $\frac{1}{2} \times \frac{1}{2}$ — type as a/b

2 $\frac{1}{2} \times \frac{1}{3}$

3 $\frac{1}{3} \times \frac{1}{3}$

4 $\frac{1}{2} \times \frac{1}{4}$

5 $\frac{1}{5} \times \frac{1}{2}$

6 $\frac{1}{4} \times \frac{1}{4}$

7 $\frac{1}{4} \times \frac{1}{2}$

8 $\frac{1}{3} \times \frac{1}{4}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

9. $\frac{2}{3} \times \frac{1}{2}$

.....

10. $\frac{3}{4} \times \frac{2}{3}$

.....

11. $\frac{2}{5} \times \frac{3}{4}$

.....

12. $\frac{5}{6} \times \frac{3}{5}$

.....

13. $\frac{3}{8} \times \frac{4}{9}$

.....

14. $\frac{2}{3} \times \frac{3}{4}$

.....

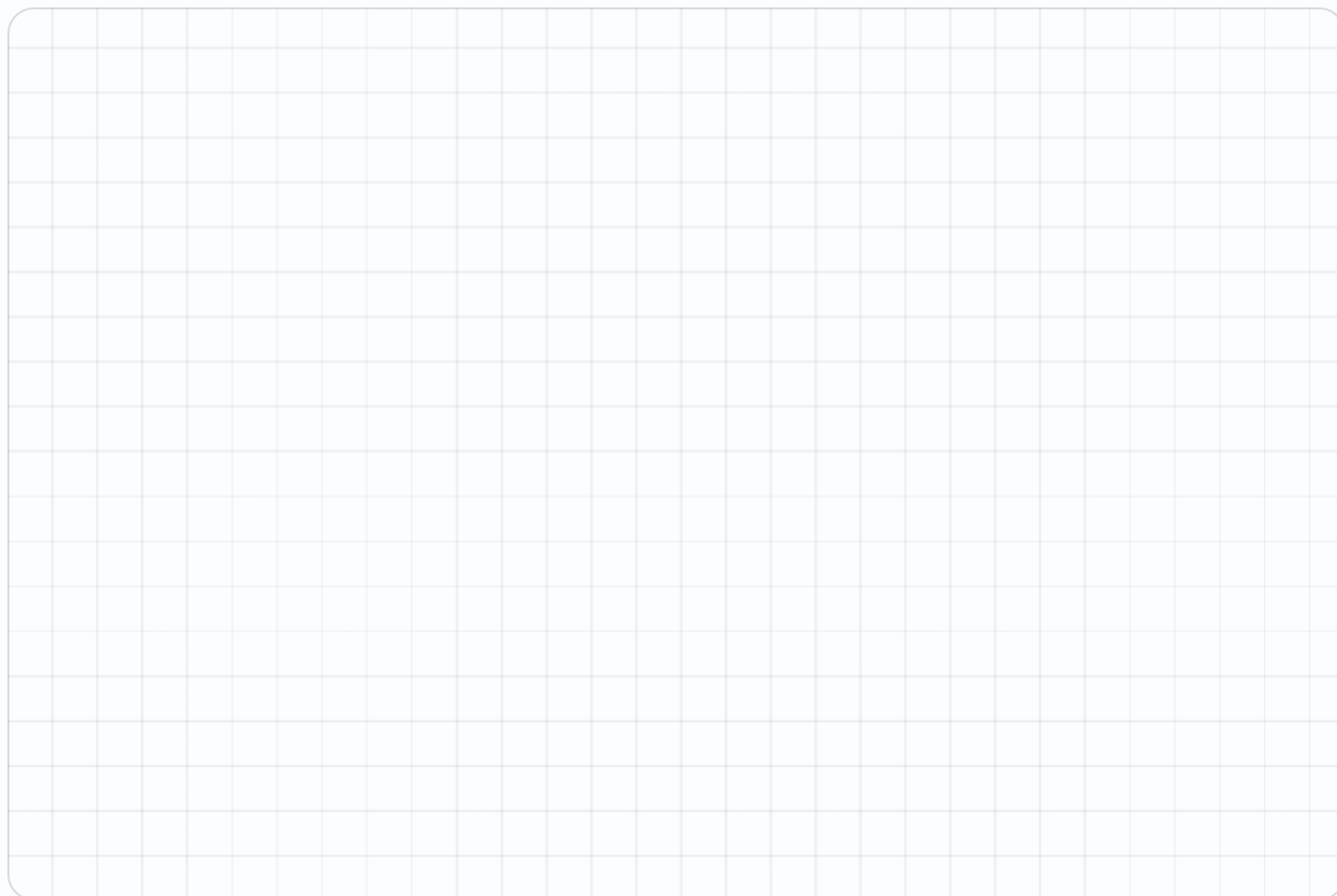
★ Challenge NOT GRADED • REASON IT OUT

19. $\frac{2}{3} \times \frac{3}{4} \times \frac{1}{2}$
.....

20. Area of a rectangle $\frac{2}{3}$ by $\frac{3}{5}$ — type as a/b
.....

21. $\frac{5}{6} \times \frac{9}{10}$
.....

22. $\frac{7}{8} \times \frac{4}{21}$
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Scaling

Predict bigger or smaller before you calculate.

Warm-Up 12 ITEMS • RECALL

1 6×2 — bigger or smaller than 6

2 $6 \times 1/2$ — bigger or smaller

3 6×1 — bigger, smaller or same

4 $6 \times 1/2$

5 $8 \times 1/4$

6 $10 \times 3/5$

7 8×1 — bigger, smaller or same

8 8×2 — bigger or smaller

9 $8 \times 1/2$ — bigger or smaller

10 $8 \times 1/2$

11 $8 \times 3/4$

12 8×2

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

13. $20 \times 3/4$ — bigger or smaller than 20

.....

14. $20 \times 5/4$ — bigger or smaller

.....

15. $20 \times 5/4$

.....

16. $12 \times 7/6$

.....

17. Why does $\times 4/4$ change nothing — type the value of $4/4$

.....

18. $20 \times 3/4$

.....



Dividing with Fractions

How many of these fit inside that?

Warm-Up 11 ITEMS • RECALL

1 How many halves in 1

2 How many halves in 3

3 How many quarters in 1

4 How many quarters in 2

5 How many thirds in 2

6 $4 \div \frac{1}{2}$

7 $3 \div \frac{1}{2}$

8 $2 \div \frac{1}{4}$

9 $5 \div \frac{1}{2}$

10 $1 \div \frac{1}{8}$

11 $6 \div \frac{1}{3}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12. $5 \div \frac{1}{4}$

13. $\frac{1}{2} \div 4$ — type as a/b

14. $\frac{1}{3} \div 2$ — type as a/b

15. $\frac{3}{4}$ as a division — type $3 \div 4$ as a decimal

16. $7 \div \frac{1}{3}$

17. $\frac{1}{3} \div 2$

Half a Garden

Enrichment. Fractions of real areas.

Warm-Up 4 ITEMS • RECALL

1 Half of a 12 m² bed
_____2 A quarter of 20 m²
_____3 $\frac{1}{2}$ of 20
_____4 $\frac{1}{4}$ of 20

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.5. $\frac{2}{3}$ of a 18 m² bed
.....6. A bed $\frac{1}{2}$ m by $\frac{3}{4}$ m — area as a/b
.....7. $\frac{2}{3}$ of 24
.....8. $\frac{3}{4} \times \frac{2}{5}$ — type as a/b
.....

Mark the reasoning, not the handwriting.

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

1.1 • Fraction of a Number

OUT OF 14

Warm-Up • 1-8 • 5 | 3 | 3 | 4 | 15 | 3 | 4 | 4

Core • 9-14 • 8 | 15 | 14 | 20 | 15 | 10

Challenge • 18-21 • 30 | 175 | 60 | 84

1.2 • Fraction × Fraction

OUT OF 14

Warm-Up • 1-8 • $\frac{1}{4}$ | $\frac{1}{6}$ | $\frac{1}{9}$ | $\frac{1}{8}$ | $\frac{1}{10}$ | $\frac{1}{16}$ | $\frac{1}{8}$ | $\frac{1}{12}$

Core • 9-14 • $\frac{1}{3}$ | $\frac{1}{2}$ | $\frac{3}{10}$ | $\frac{1}{2}$ | $\frac{1}{6}$ | $\frac{1}{2}$

Challenge • 19-22 • $\frac{1}{4}$ | $\frac{2}{5}$ | $\frac{3}{4}$ | $\frac{1}{6}$

1.3 • Scaling

OUT OF 18

Warm-Up • 1-12 • bigger | smaller | same | 3 | 2 | 6 | same | bigger | smaller | 4 | 6 | 16

Core • 13-18 • smaller | bigger | 25 | 14 | 1 | 15

Challenge • 19-22 • $\frac{4}{3}$ | 30 | smaller | 20

1.4 • Dividing with Fractions

OUT OF 17

Warm-Up • 1-11 • 2 | 6 | 4 | 8 | 6 | 8 | 6 | 8 | 10 | 8 | 18

Core • 12-17 • 20 | $\frac{1}{8}$ | $\frac{1}{6}$ | 0.75 | 21 | $\frac{1}{6}$

Challenge • 21-24 • 12 | $\frac{1}{12}$ | 24 | $\frac{1}{24}$

Fri • Half a Garden

OUT OF 8

Warm-Up • 1-4 • 6 | 5 | 10 | 5

Core • 5-8 • 12 | $\frac{3}{8}$ | 16 | $\frac{3}{10}$

Challenge • 9-14 • 27 | $\frac{1}{2}$ | 12 | $\frac{3}{8}$ | 15 | 24

Fold It Twice

Half of a third is a sixth. Do it on paper before you do it on paper.

Warm-Up 12 ITEMS • RECALL

1 $\frac{1}{2} \times \frac{1}{2}$

2 $\frac{1}{2} \times \frac{1}{3}$

3 $\frac{1}{3} \times \frac{1}{3}$

4 $\frac{1}{2} \times \frac{1}{4}$

5 $\frac{1}{5} \times \frac{1}{2}$

6 $\frac{1}{4} \times \frac{1}{4}$

7 $\frac{1}{2}$ of $\frac{1}{2}$ — type as a/b

8 $\frac{1}{2}$ of $\frac{1}{4}$

9 $\frac{1}{3}$ of $\frac{1}{2}$

10 $\frac{1}{2}$ of $\frac{1}{3}$

11 $\frac{1}{4}$ of $\frac{1}{4}$

12 $\frac{1}{2}$ of $\frac{1}{5}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $\frac{1}{2} \times \frac{1}{5}$

14. $\frac{1}{3} \times \frac{1}{4}$

15. $\frac{1}{6} \times \frac{1}{2}$

16. $\frac{1}{3} \times \frac{1}{5}$

17. $\frac{1}{4} \times \frac{1}{5}$

18. $\frac{1}{3}$ of $\frac{3}{4}$

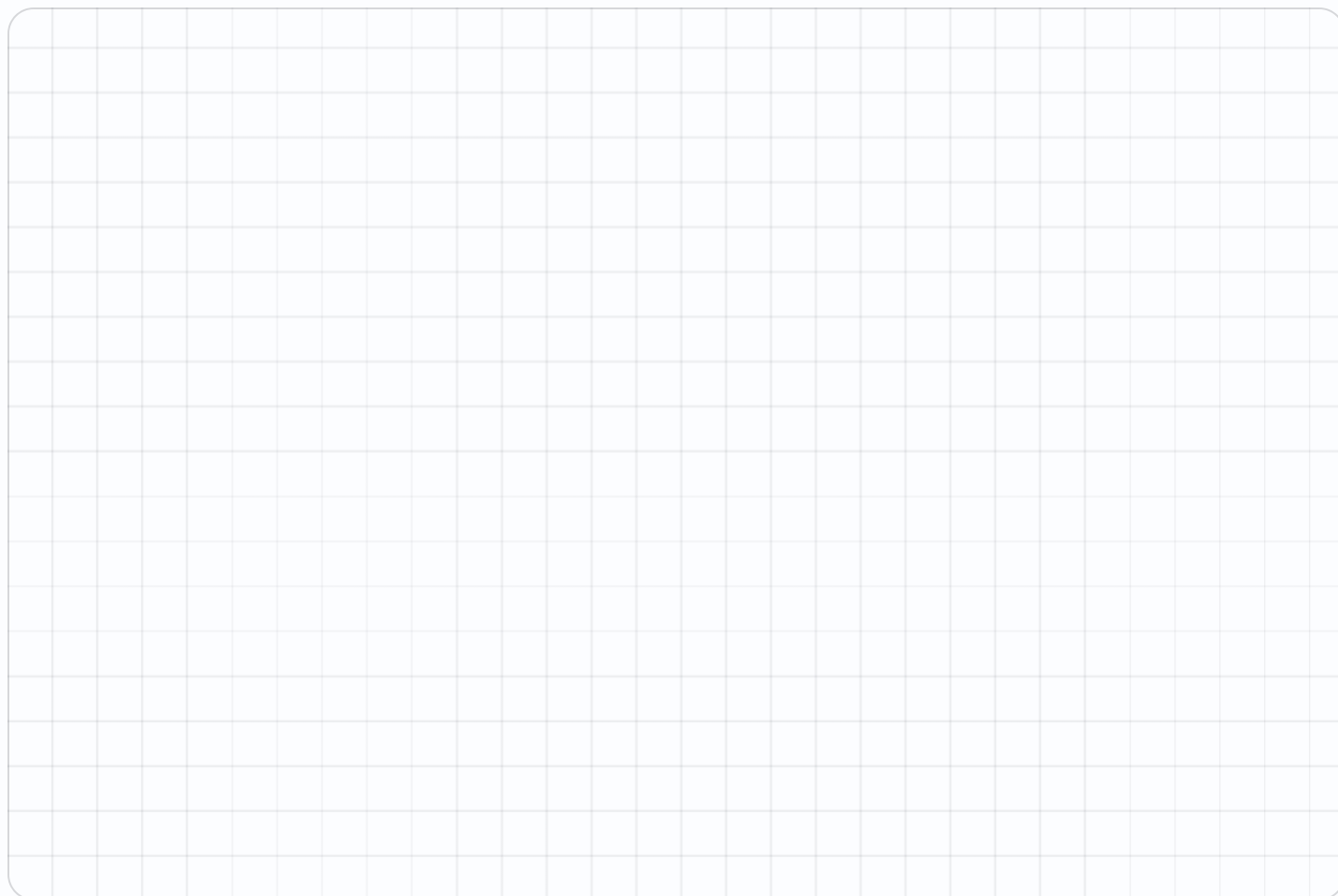
★ Challenge NOT GRADED • REASON IT OUT

23. $\frac{1}{2} \times \frac{1}{3} \times \frac{1}{4}$
.....

24. $\frac{1}{8} \times \frac{1}{8}$
.....

25. $\frac{2}{3}$ of $\frac{3}{8}$
.....

26. $\frac{5}{6}$ of $\frac{3}{10}$
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

The Area Model

A rectangle cut both ways. The overlap is the answer.

Warm-Up 12 ITEMS • RECALL

1 A $\frac{1}{2}$ by $\frac{1}{2}$ square — its area

2 A $\frac{1}{2}$ by $\frac{1}{3}$ rectangle

3 A $\frac{1}{3}$ by $\frac{1}{4}$ rectangle

4 Pieces when a square is cut in halves both ways

5 Cut in thirds both ways

6 Cut in halves and thirds

7 A 1 by 2 rectangle — area

8 A 2 by 3 rectangle

9 A $\frac{1}{2}$ by 1 strip — type as a/b

10 A $\frac{1}{2}$ by $\frac{1}{2}$ square

11 A $\frac{1}{3}$ by 1 strip

12 A $\frac{1}{4}$ by $\frac{1}{2}$ rectangle

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. A $\frac{2}{3}$ by $\frac{1}{2}$ rectangle
.....

14. A $\frac{3}{4}$ by $\frac{2}{3}$ rectangle
.....

15. A $\frac{2}{5}$ by $\frac{3}{4}$ rectangle
.....

16. A $\frac{2}{3}$ by $\frac{3}{5}$ rectangle
.....

17. A $\frac{5}{6}$ by $\frac{3}{5}$ rectangle
.....

18. A $\frac{2}{3}$ by $\frac{3}{4}$ rectangle — type as a/b
.....

★ Challenge NOT GRADED • REASON IT OUT23. A $\frac{3}{8}$ by $\frac{4}{9}$ rectangle24. A $\frac{2}{3}$ by $\frac{3}{4}$ by $\frac{1}{2}$ box — its volume25. A $\frac{7}{8}$ by $\frac{4}{7}$ rectangle26. A $\frac{5}{6}$ by $\frac{9}{10}$ rectangle**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Multiply Across

Numerators, then denominators. Then simplify.

✦ Warm-Up 9 ITEMS • RECALL

1 $1/2 \times 2/3$

2 $2/3 \times 1/2$

3 $1/4 \times 2/3$

4 $3/4 \times 1/3$

5 $2/5 \times 1/2$

6 $1/3 \times 3/4$

7 $1/2 \times 1/3$ — type as a/b

8 $1/4 \times 1/5$

9 $1/2 \times 4/5$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

10. $3/4 \times 2/3$

11. $2/5 \times 3/4$

12. $5/6 \times 3/5$

13. $3/8 \times 4/9$

14. $4/5 \times 5/8$

15. $3/4 \times 5/6$

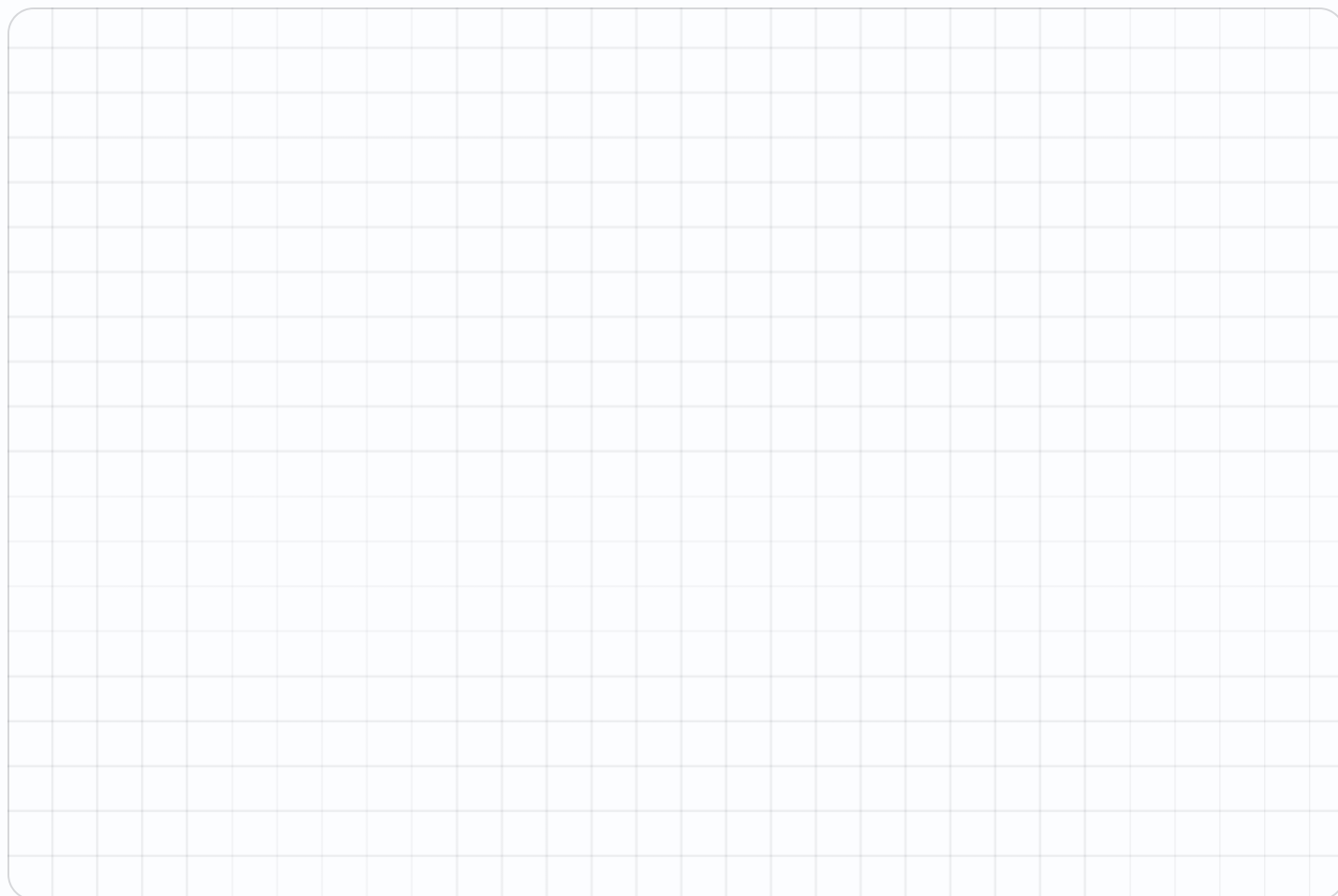
★ Challenge NOT GRADED • REASON IT OUT

20. $\frac{2}{3} \times \frac{3}{4} \times \frac{1}{2}$
.....

21. $\frac{6}{7} \times \frac{7}{12}$
.....

22. $\frac{9}{16} \times \frac{8}{27}$
.....

23. $\frac{11}{12} \times \frac{6}{11}$
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Simplify as You Go

Cancel before multiplying and the numbers stay small.

Warm-Up 12 ITEMS • RECALL

1 $2/4$ simplified2 $3/6$ 3 $4/8$ 4 $6/9$ 5 $5/10$ 6 $8/12$ 7 $2/4$ simplified — type as a/b8 $3/6$ simplified9 $4/6$ simplified10 $6/8$ simplified11 $5/10$ simplified12 $9/12$ simplified

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $2/3 \times 3/8$ — simplified

.....

14. $4/9 \times 3/8$

.....

15. $5/12 \times 6/10$

.....

16. $9/10 \times 5/6$

.....

17. $7/8 \times 4/7$

.....

18. $2/3 \times 3/5$ simplified

.....

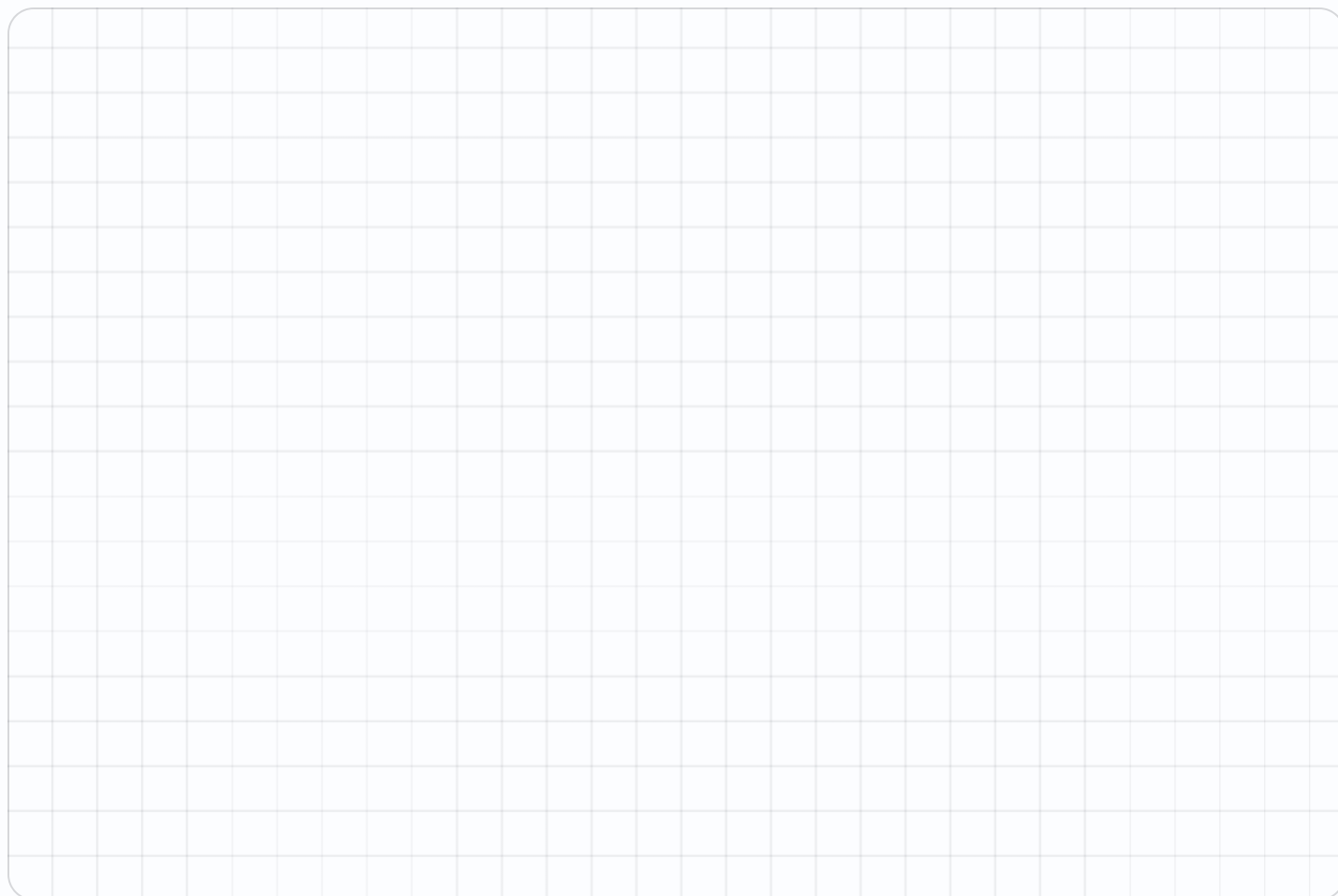
★ Challenge NOT GRADED • REASON IT OUT

23. $12/25 \times 5/6$
.....

24. $15/16 \times 8/45$
.....

25. $12/25 \times 5/6$ simplified
.....

26. $15/16 \times 8/45$ simplified
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Scaling Showdown

Bigger, smaller or the same — predict before you calculate.

Warm-Up 4 ITEMS • RECALL

1 6×2 — bigger or smaller than 6

2 $6 \times \frac{1}{2}$ — bigger or smaller

3 $\frac{1}{2}$ of $\frac{1}{2}$ — type as a/b

4 $\frac{1}{2} \times 6$

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. $20 \times \frac{3}{4}$ — bigger or smaller

.....

6. $20 \times \frac{5}{4}$ — bigger or smaller

.....

7. $\frac{2}{3} \times \frac{3}{4}$ — type as a/b

.....

8. $\frac{3}{5} \times 10$

.....

Week 2 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

2.1 • Fold It Twice

OUT OF 18

Warm-Up • 1-12 • $\frac{1}{4}$ | $\frac{1}{6}$ | $\frac{1}{9}$ | $\frac{1}{8}$ | $\frac{1}{10}$ | $\frac{1}{16}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{16}$ | $\frac{1}{10}$

Core • 13-18 • $\frac{1}{10}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{15}$ | $\frac{1}{20}$ | $\frac{1}{4}$

Challenge • 23-26 • $\frac{1}{24}$ | $\frac{1}{64}$ | $\frac{1}{4}$ | $\frac{1}{4}$

2.2 • The Area Model

OUT OF 18

Warm-Up • 1-12 • $\frac{1}{4}$ | $\frac{1}{6}$ | $\frac{1}{12}$ | 4 | 9 | 6 | 2 | 6 | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{3}$ | $\frac{1}{8}$

Core • 13-18 • $\frac{1}{3}$ | $\frac{1}{2}$ | $\frac{3}{10}$ | $\frac{2}{5}$ | $\frac{1}{2}$ | $\frac{1}{2}$

Challenge • 23-26 • $\frac{1}{6}$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{3}{4}$

2.3 • Multiply Across

OUT OF 15

Warm-Up • 1-9 • $\frac{1}{3}$ | $\frac{1}{3}$ | $\frac{1}{6}$ | $\frac{1}{4}$ | $\frac{1}{5}$ | $\frac{1}{4}$ | $\frac{1}{6}$ | $\frac{1}{20}$ | $\frac{2}{5}$

Core • 10-15 • $\frac{1}{2}$ | $\frac{3}{10}$ | $\frac{1}{2}$ | $\frac{1}{6}$ | $\frac{1}{2}$ | $\frac{5}{8}$

Challenge • 20-23 • $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{6}$ | $\frac{1}{2}$

2.4 • Simplify as You Go

OUT OF 18

Warm-Up • 1-12 • $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{3}{4}$

Core • 13-18 • $\frac{1}{4}$ | $\frac{1}{6}$ | $\frac{1}{4}$ | $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{2}{5}$

Challenge • 23-26 • $\frac{2}{5}$ | $\frac{1}{6}$ | $\frac{2}{5}$ | $\frac{1}{6}$

Fri • Scaling Showdown

OUT OF 8

Warm-Up • 1-4 • bigger | smaller | $\frac{1}{4}$ | 3

Core • 5-8 • smaller | bigger | $\frac{1}{2}$ | 6

Challenge • 9-14 • same | $\frac{4}{3}$ | 30 | $\frac{3}{4}$ | 14 | $\frac{1}{2}$

Bigger or Smaller

Decide without calculating. Then check yourself.

Warm-Up 9 ITEMS • RECALL

1 10×2 — bigger or smaller than 10

2 $10 \times 1/2$

3 10×1 — bigger, smaller or same

4 $8 \times 1/4$ — bigger or smaller

5 8×3 — bigger or smaller

6 $8 \times 1/2$

7 $10 \times 1/2$ — bigger or smaller

8 $10 \times 3/2$

9 $10 \times 1/5$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. $24 \times 2/3$ — bigger or smaller

.....

11. $24 \times 2/3$

.....

12. $24 \times 3/2$ — bigger or smaller

.....

13. $24 \times 3/2$

.....

14. $12 \times 7/6$

.....

15. $24 \times 2/3$ — bigger or smaller than 24

.....



Why $\times 1$ Changes Nothing

$3/3$ is one wearing a costume.

Warm-Up 12 ITEMS • RECALL

1 $3/3$ as a whole number

2 $5/5$

3 $8/8$

4 12×1

5 $12 \times 3/3$

6 $7 \times 4/4$

7 5×1

8 $5 \times 2/2$

9 $5 \times 3/3$

10 $2/2$ as a whole number

11 $8/8$ as a whole number

12 $12 \times 4/4$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $20 \times 5/5$

14. $2/3 \times 4/4$ — type as a/b

15. $8/12$ simplified

16. $15 \times 6/6$

17. $3/4 \times 5/5$ — type as a/b

18. $3/4 \times 4/4$ — type as a/b

Compare Products

Which is larger, without working either one out.

Warm-Up 12 ITEMS • RECALL

1 Larger: 10×2 or 10×3 — type the multiplier

2 Larger: $10 \times \frac{1}{2}$ or $10 \times \frac{1}{3}$

3 Larger: 8×1 or 8×2

4 Larger: $6 \times \frac{1}{4}$ or $6 \times \frac{1}{2}$

5 $10 \times \frac{1}{2}$

6 $10 \times \frac{1}{3}$ — bigger or smaller than 5

7 Bigger — type 8×2 or $8 \times \frac{1}{2}$

8 Bigger — type 10×1 or $10 \times \frac{3}{4}$

9 $8 \times \frac{1}{2}$

10 8×2

11 $10 \times \frac{3}{4}$

12 $12 \times \frac{1}{3}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Larger: $20 \times \frac{3}{4}$ or $20 \times \frac{4}{5}$ — type the fraction

.....

14. Larger: $12 \times \frac{5}{6}$ or $12 \times \frac{7}{8}$

.....

15. $20 \times \frac{4}{5}$

.....

16. $12 \times \frac{7}{8}$

.....

17. Larger: $30 \times \frac{2}{3}$ or $30 \times \frac{3}{5}$

.....

18. Without computing, bigger — type $20 \times \frac{4}{5}$ or $20 \times \frac{6}{5}$

.....



Working space



Error journal

Scaling in Context

Recipes, distances and prices, scaled up and down.

Warm-Up 11 ITEMS • RECALL

1 Half of 20 km

2 Double 15 km

3 $\frac{1}{4}$ of \$40

4 $\frac{3}{4}$ of 20

5 $\frac{1}{2}$ of \$18

6 $\frac{2}{3}$ of 12

7 $\frac{1}{2}$ of a 20 m plot

8 $\frac{1}{4}$ of 40 m

9 $\frac{2}{3}$ of 30

10 $\frac{1}{5}$ of 50

11 $\frac{1}{2}$ of 8 hours

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12. $\frac{3}{4}$ of a \$60 price

.....

13. A $\frac{2}{3}$ scaled recipe from 18 cups

.....

14. $\frac{5}{6}$ of a 24 km route

.....

15. A price of \$80 at $\frac{3}{5}$

.....

16. $\frac{7}{8}$ of 32

.....

17. A recipe scaled by $\frac{3}{4}$ from 20 cups

.....



Working space



Error journal

Measure the Garden

Real beds, real fractions.

✦ Warm-Up 4 ITEMS • RECALL

1 Half of a 12 m² bed

2 A quarter of 20 m²

3 A 4 by 5 garden — area

4 Half of it

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. $\frac{2}{3}$ of an 18 m² bed

.....

6. A bed $\frac{1}{2}$ m by $\frac{3}{4}$ m — area

.....

7. A $\frac{1}{2}$ by $\frac{3}{4}$ plot — type as a/b

.....

8. $\frac{3}{8}$ of a 40 m² garden

.....

 Challenge NOT GRADED • REASON IT OUT

9. $\frac{3}{5}$ of a 45 m^2 plot

10. A $\frac{2}{3}$ m by $\frac{3}{4}$ m bed — area

11. How many $\frac{1}{4} \text{ m}^2$ tiles cover 3 m^2

12. A $\frac{2}{3}$ by $\frac{5}{6}$ plot — type as a/b

13. $\frac{5}{9}$ of a 63 m^2 garden

14. A $\frac{3}{4}$ by $\frac{4}{5}$ plot as a fraction of the whole

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 • Bigger or Smaller

OUT OF 15

Warm-Up • 1-9 • bigger | smaller | same | smaller | bigger | 4 | smaller | 15 | 2

Core • 10-15 • smaller | 16 | bigger | 36 | 14 | smaller

Challenge • 19-22 • smaller | 36 | same | 35

3.2 • Why $\times 1$ Changes Nothing

OUT OF 18

Warm-Up • 1-12 • 1 | 1 | 1 | 12 | 12 | 7 | 5 | 5 | 5 | 1 | 1 | 12

Core • 13-18 • 20 | $8/12$ | $2/3$ | 15 | $15/20$ | $3/4$

Challenge • 23-26 • same | 9 | $7/8$ | 1

3.3 • Compare Products

OUT OF 18

Warm-Up • 1-12 • 3 | $1/2$ | 2 | $1/2$ | 5 | smaller | 8×2 | 10×1 | 4 | 16 | 7.5 | 4

Core • 13-18 • $4/5$ | $7/8$ | 16 | 10.5 | $2/3$ | $20 \times 6/5$

Challenge • 22-25 • $5/8$ | 15 | $50 \times 11/10$ | 10

3.4 • Scaling in Context

OUT OF 17

Warm-Up • 1-11 • 10 | 30 | 10 | 15 | 9 | 8 | 10 | 10 | 20 | 10 | 4

Core • 12-17 • 45 | 12 | 20 | 48 | 28 | 15

Challenge • 22-25 • 200 | 36 | $1/2$ | 3

Fri • Measure the Garden

OUT OF 8

Warm-Up • 1-4 • 6 | 5 | 20 | 10

Core • 5-8 • 12 | $3/8$ | $3/8$ | 15

Challenge • 9-14 • 27 | $1/2$ | 12 | $5/9$ | 35 | $3/5$

Whole ÷ Unit Fraction

How many of these fit inside that?

Warm-Up 12 ITEMS • RECALL

1 How many halves in 1

2 How many halves in 3

3 How many quarters in 1

4 How many quarters in 2

5 How many thirds in 2

6 $4 \div \frac{1}{2}$

7 $2 \div \frac{1}{2}$

8 $3 \div \frac{1}{3}$

9 $4 \div \frac{1}{4}$

10 $5 \div \frac{1}{5}$

11 $1 \div \frac{1}{6}$

12 $2 \div \frac{1}{4}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $5 \div \frac{1}{4}$

.....

14. $7 \div \frac{1}{3}$

.....

15. $3 \div \frac{1}{8}$

.....

16. $6 \div \frac{1}{5}$

.....

17. $10 \div \frac{1}{2}$

.....

18. $6 \div \frac{1}{3}$

.....



Working space



Error journal

Unit Fraction ÷ Whole

Sharing a third between four people.

Warm-Up 7 ITEMS • RECALL

1 $\frac{1}{2} \div 2$

2 $\frac{1}{2} \div 3$

3 $\frac{1}{3} \div 2$

4 $\frac{1}{4} \div 2$

5 $\frac{1}{5} \div 2$

6 $\frac{1}{3} \div 3$

7 $\frac{1}{2} \div 2$ — type as a/b

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

8. $\frac{1}{2} \div 4$

9. $\frac{1}{3} \div 4$

10. $\frac{1}{4} \div 3$

11. $\frac{1}{5} \div 4$

12. $\frac{1}{6} \div 2$

13. $\frac{1}{2} \div 5$ — type as a/b



Working space



Error journal

Fraction as Division

$\frac{3}{4}$ is literally $3 \div 4$.

Warm-Up 12 ITEMS • RECALL

1 $1 \div 2$ as a decimal

2 $1 \div 4$ as a decimal

3 $3 \div 4$ as a decimal

4 $1 \div 5$ as a decimal

5 $2 \div 5$ as a decimal

6 $1 \div 10$ as a decimal

7 $6 \div 2$ as a fraction — type as a/b

8 $3 \div 4$ — type as a/b

9 $1 \div 2$

10 $5 \div 8$

11 $2 \div 3$

12 $7 \div 10$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $3 \div 8$ as a decimal

.....

14. $5 \div 8$ as a decimal

.....

15. $7 \div 10$ as a decimal

.....

16. 3 pizzas between 4 — each, as a/b

.....

17. 5 cakes between 2 — each, as a/b

.....

18. 3 pizzas shared by 4 — each share as a/b

.....

Ribbon, Pizza and Time

Real contexts for dividing with fractions.

Warm-Up 12 ITEMS • RECALL

1 A 2 m ribbon into $\frac{1}{2}$ m pieces

2 A 3 m ribbon into $\frac{1}{2}$ m pieces

3 Half a pizza between 2

4 A whole pizza in quarters — slices

5 An hour in half hours

6 An hour in quarter hours

7 A 6 m ribbon into $\frac{1}{2}$ m pieces

8 A 4 m ribbon into $\frac{1}{4}$ m pieces

9 A pizza in $\frac{1}{8}$ slices — slices

10 2 pizzas in $\frac{1}{8}$ slices

11 3 hours in $\frac{1}{2}$ hours

12 A 5 m ribbon into 1 m pieces

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. A 4.5 m rope into $\frac{1}{2}$ m pieces

.....

14. $\frac{1}{3}$ of a cake between 2

.....

15. 2 hours in $\frac{1}{4}$ hours

.....

16. A 5 m ribbon into $\frac{1}{4}$ m bows

.....

17. $\frac{1}{2}$ of a pie between 3

.....

18. A 7 m ribbon into $\frac{1}{4}$ m pieces

.....



Working space



Error journal

Mid-Unit Quiz

Eight items across Weeks 1–4. 85% to keep flying.

Warm-Up 4 ITEMS • RECALL

1 $\frac{1}{2} \times \frac{1}{3}$

2 $\frac{2}{3}$ of 9

3 $\frac{1}{2}$ of 12

4 $4 \div \frac{1}{2}$

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. $\frac{3}{4} \times \frac{2}{3}$

6. $24 \times \frac{2}{3}$

7. $5 \div \frac{1}{4}$

8. $\frac{1}{3} \div 4$

9. $20 \times \frac{3}{4}$ — bigger or smaller than 20

10. $\frac{2}{3} \times \frac{3}{4}$ — type as a/b

11. $\frac{1}{2} \div 4$

12. $\frac{3}{4}$ of 20

13. $6 \div \frac{1}{3}$

14. $3 \div 4$ — type as a/b

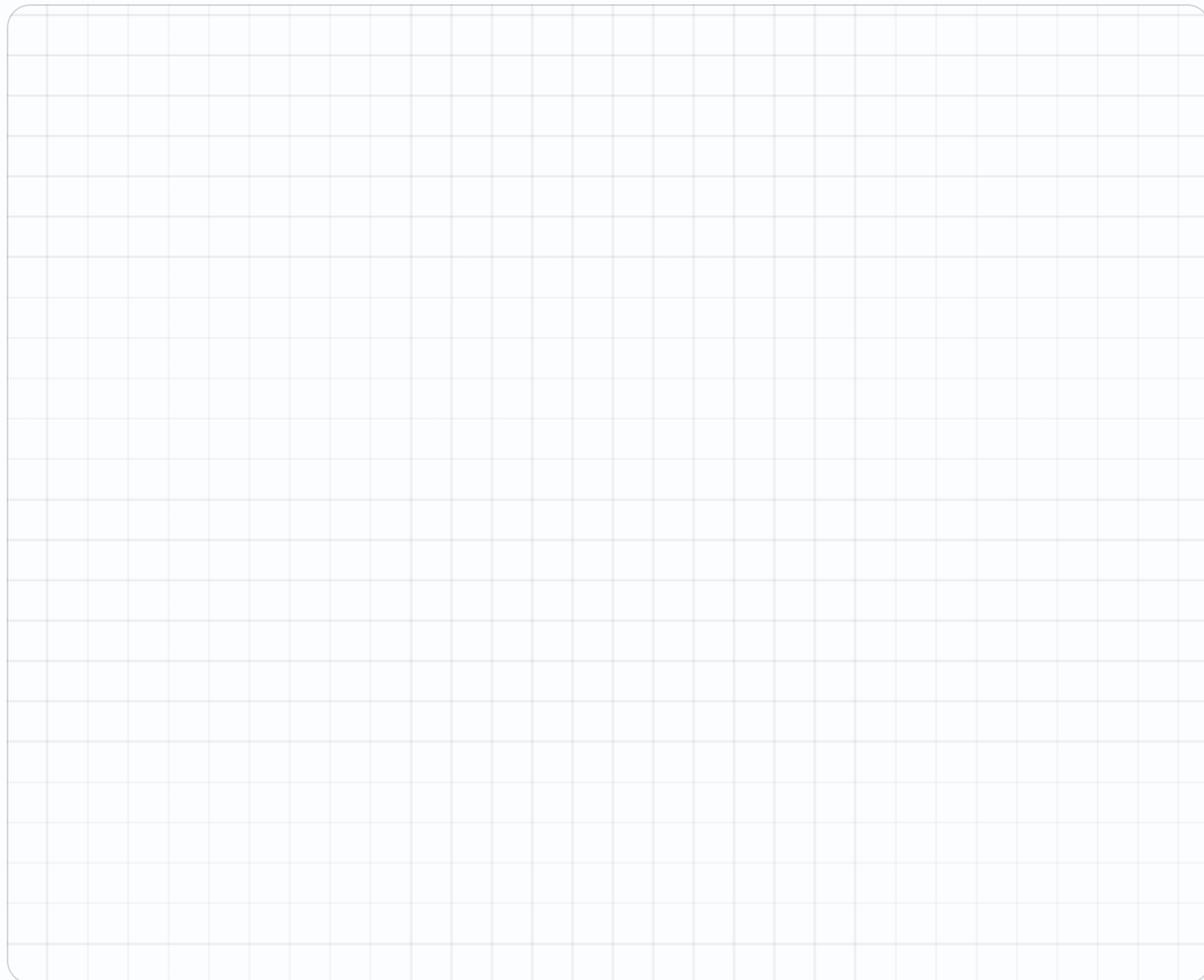
★ Challenge NOT GRADED • REASON IT OUT

15. $\frac{2}{3} \times \frac{3}{4} \times \frac{1}{2}$

.....

16. $\frac{5}{6} \times \frac{9}{10}$

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Whole ÷ Unit Fraction

OUT OF 18

Warm-Up • 1-12 • 2 | 6 | 4 | 8 | 6 | 8 | 4 | 9 | 16 | 25 | 6 | 8

Core • 13-18 • 20 | 21 | 24 | 30 | 20 | 18

Challenge • 22-25 • 12 | 72 | 40 | 36

4.2 • Unit Fraction ÷ Whole

OUT OF 13

Warm-Up • 1-7 • $\frac{1}{4}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{8}$ | $\frac{1}{10}$ | $\frac{1}{9}$ | $\frac{1}{4}$

Core • 8-13 • $\frac{1}{8}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{20}$ | $\frac{1}{12}$ | $\frac{1}{10}$

Challenge • 18-21 • $\frac{1}{12}$ | $\frac{1}{32}$ | $\frac{1}{30}$ | $\frac{1}{12}$

4.3 • Fraction as Division

OUT OF 18

Warm-Up • 1-12 • 0.5 | 0.25 | 0.75 | 0.2 | 0.4 | 0.1 | 3 | $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{5}{8}$ | $\frac{2}{3}$ | $\frac{7}{10}$

Core • 13-18 • 0.375 | 0.625 | 0.7 | $\frac{3}{4}$ | $\frac{5}{2}$ | $\frac{3}{4}$

Challenge • 23-26 • 0.875 | 2.25 | 3 | 2.6

4.4 • Ribbon, Pizza and Time

OUT OF 18

Warm-Up • 1-12 • 4 | 6 | $\frac{1}{4}$ | 4 | 2 | 4 | 12 | 16 | 8 | 16 | 6 | 5

Core • 13-18 • 9 | $\frac{1}{6}$ | 8 | 20 | $\frac{1}{6}$ | 28

Challenge • 23-26 • 24 | 18 | $\frac{1}{15}$ | 8

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • $\frac{1}{6}$ | 6 | 6 | 8

Core • 5-14 • $\frac{1}{2}$ | 16 | 20 | $\frac{1}{12}$ | smaller | $\frac{1}{2}$ | $\frac{1}{8}$ | 15 | 18 | $\frac{3}{4}$

Challenge • 15-16 • $\frac{1}{4}$ | $\frac{3}{4}$

Finish the Garden Plan

Every bed with its fraction and its area.

Warm-Up 12 ITEMS • RECALL

1 $\frac{1}{2}$ of 20 m²

2 $\frac{1}{4}$ of 24 m²

3 $\frac{1}{3}$ of 27 m²

4 $\frac{2}{3}$ of 12 m²

5 $\frac{3}{4}$ of 16 m²

6 $\frac{1}{5}$ of 30 m²

7 A $\frac{1}{2}$ by $\frac{1}{2}$ plot — type as a/b

8 $\frac{1}{4}$ of a 40 m² garden

9 $\frac{1}{2}$ of 30 m²

10 A 2 by 3 bed — area

11 $\frac{1}{3}$ of 24 m²

12 $\frac{3}{4}$ of 12 m²

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $\frac{2}{5}$ of 45 m²

.....

14. $\frac{5}{6}$ of 36 m²

.....

15. A $\frac{3}{4}$ m by $\frac{2}{3}$ m bed — area

.....

16. $\frac{3}{8}$ of 64 m²

.....

17. $\frac{7}{10}$ of 50 m²

.....

18. A $\frac{2}{3}$ by $\frac{3}{4}$ plot — type as a/b

.....

★ Challenge NOT GRADED • REASON IT OUT

23. A $\frac{2}{3}$ m by $\frac{5}{6}$ m bed — area

.....

24. How many $\frac{1}{8}$ m² tiles cover 2 m²

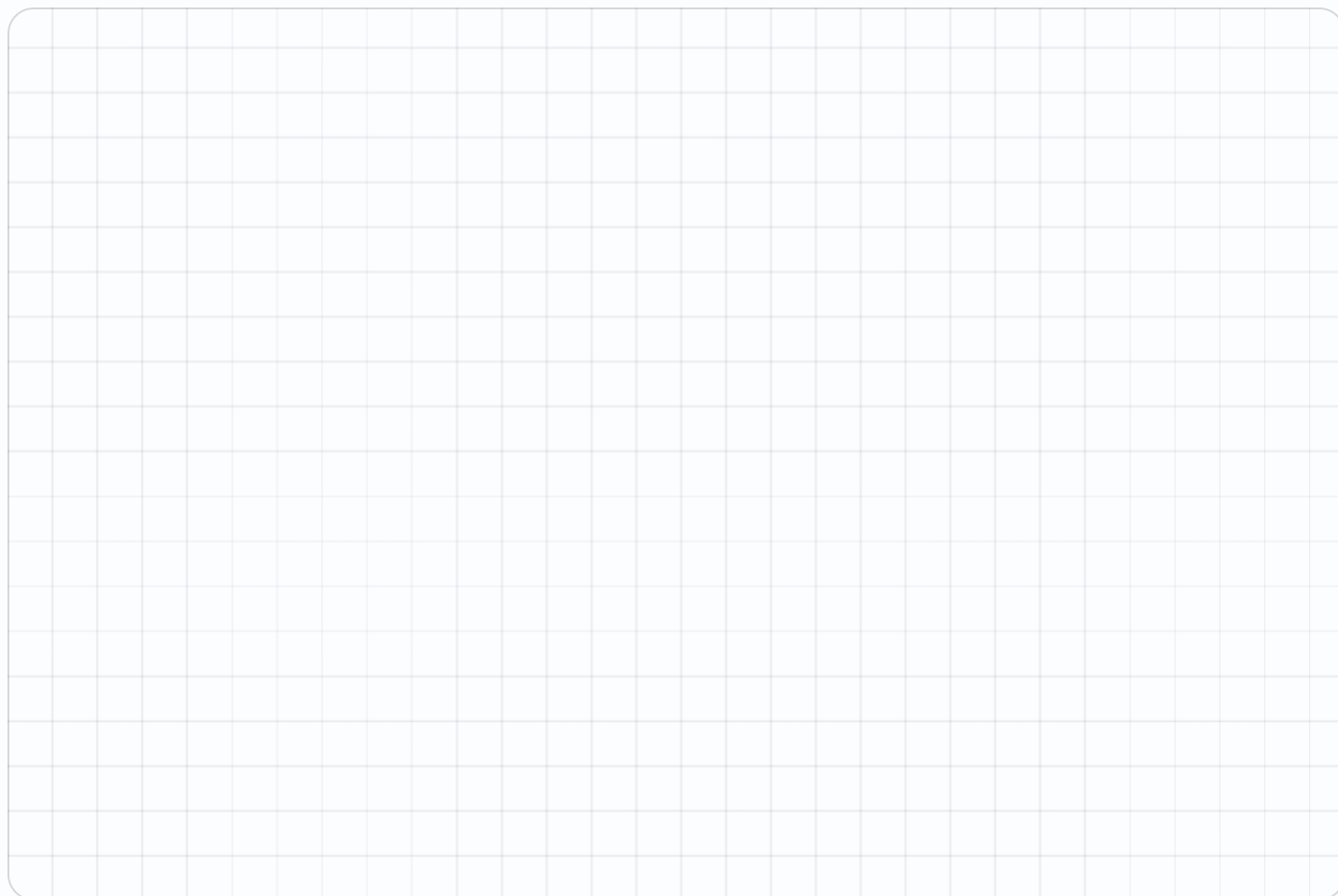
.....

25. A $\frac{5}{8}$ by $\frac{4}{5}$ plot as a/b

.....

26. Three beds each $\frac{2}{9}$ of a 54 m² garden — total m²

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Explain a Shrink

Name one product smaller than its starting number and say why.

Warm-Up 11 ITEMS • RECALL

1 $12 \times \frac{1}{2}$

2 $12 \times \frac{1}{3}$

3 $12 \times \frac{1}{4}$

4 12×2

5 12×1

6 $12 \times \frac{3}{4}$

7 $20 \times \frac{1}{2}$

8 $20 \times \frac{1}{4}$

9 Multiplying by $\frac{1}{2}$ makes it — type half or double

10 $30 \times \frac{1}{5}$

11 $16 \times \frac{1}{8}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Which shrinks 12: $\times \frac{3}{4}$ or $\times \frac{5}{4}$ — type the fraction
.....

13. $30 \times \frac{2}{5}$
.....

14. $30 \times \frac{5}{2}$
.....

15. Which grows 30 — type $\frac{2}{5}$ or $\frac{5}{2}$
.....

16. $40 \times \frac{7}{8}$
.....

17. $40 \times \frac{3}{4}$ — bigger or smaller than 40
.....



Working space



Error journal

Multiply and Divide Together

Mixed set. Read before you reach for a method.

Warm-Up 10 ITEMS • RECALL

1 $\frac{1}{2} \times \frac{1}{2}$

2 $2 \div \frac{1}{2}$

3 $\frac{1}{2}$ of 8

4 $\frac{1}{3} \div 2$

5 $\frac{1}{4} \times \frac{1}{2}$

6 $3 \div \frac{1}{3}$

7 $\frac{1}{2}$ of 16

8 $4 \div \frac{1}{2}$

9 $\frac{1}{2} \times \frac{1}{4}$ — type as a/b

10 $\frac{2}{3}$ of 9

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. $\frac{3}{4} \times \frac{2}{3}$

12. $5 \div \frac{1}{4}$

13. $\frac{2}{5}$ of 35

14. $\frac{1}{4} \div 3$

15. $24 \times \frac{5}{6}$

16. $\frac{3}{4} \times \frac{2}{3}$ — type as a/b

★ Challenge NOT GRADED • REASON IT OUT

20. $\frac{2}{3} \times \frac{3}{4} \times \frac{1}{2}$
.....

21. A 3 m ribbon into $\frac{1}{4}$ m pieces
.....

22. $\frac{5}{6} \times \frac{9}{10}$
.....

23. How many $\frac{3}{4}$ m pieces from a 6 m ribbon
.....**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

Warm-Up 11 ITEMS • RECALL

1 $\frac{1}{2} \times \frac{1}{4}$

2 $\frac{1}{3}$ of 12

3 $4 \div \frac{1}{2}$

4 $\frac{1}{2} \div 2$

5 $\frac{3}{4}$ of 20

6 $\frac{1}{5} \times \frac{1}{2}$

7 $\frac{1}{2}$ of 10

8 $\frac{1}{2} \times \frac{1}{2}$ — type as a/b

9 $4 \div \frac{1}{4}$

10 $\frac{3}{4}$ of 8

11 $2 \div \frac{1}{2}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12. $\frac{2}{5} \times \frac{3}{4}$

13. $7 \div \frac{1}{3}$

14. $\frac{5}{6}$ of 24

15. $\frac{1}{5} \div 4$

16. $24 \times \frac{3}{2}$

17. $\frac{2}{3} \times \frac{3}{5}$ — type as a/b

18. $7 \div \frac{1}{4}$

19. $\frac{1}{6} \div 3$

20. $\frac{3}{8}$ of 32

21. $24 \times \frac{2}{3}$



Mission 06 Test

Twelve items plus the Big Question, answered out loud.

NAME _____
DATE _____
RIGHT ____ OF ____ TRIED

 Warm-Up 4 ITEMS • RECALL

1 $\frac{1}{2} \times \frac{1}{3}$

2 $\frac{1}{4}$ of 20

3 $\frac{1}{2}$ of 14

4 $3 \div \frac{1}{2}$

 Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. $\frac{2}{3}$ of 12

6. $\frac{3}{4} \times \frac{2}{3}$

7. $\frac{2}{5} \times \frac{3}{4}$

8. $24 \times \frac{2}{3}$

9. $20 \times \frac{5}{4}$

10. $5 \div \frac{1}{4}$

11. $\frac{1}{3} \div 4$

12. $3 \div 4$ as a decimal

13. $\frac{1}{2} \times \frac{2}{3}$ — type as a/b

14. $\frac{2}{5}$ of 45

15. $9 \div \frac{1}{3}$

16. $\frac{1}{4} \div 5$

17. $\frac{3}{4} \times \frac{8}{9}$

18. $30 \times \frac{4}{5}$

19. $5 \div 8$ — type as a/b

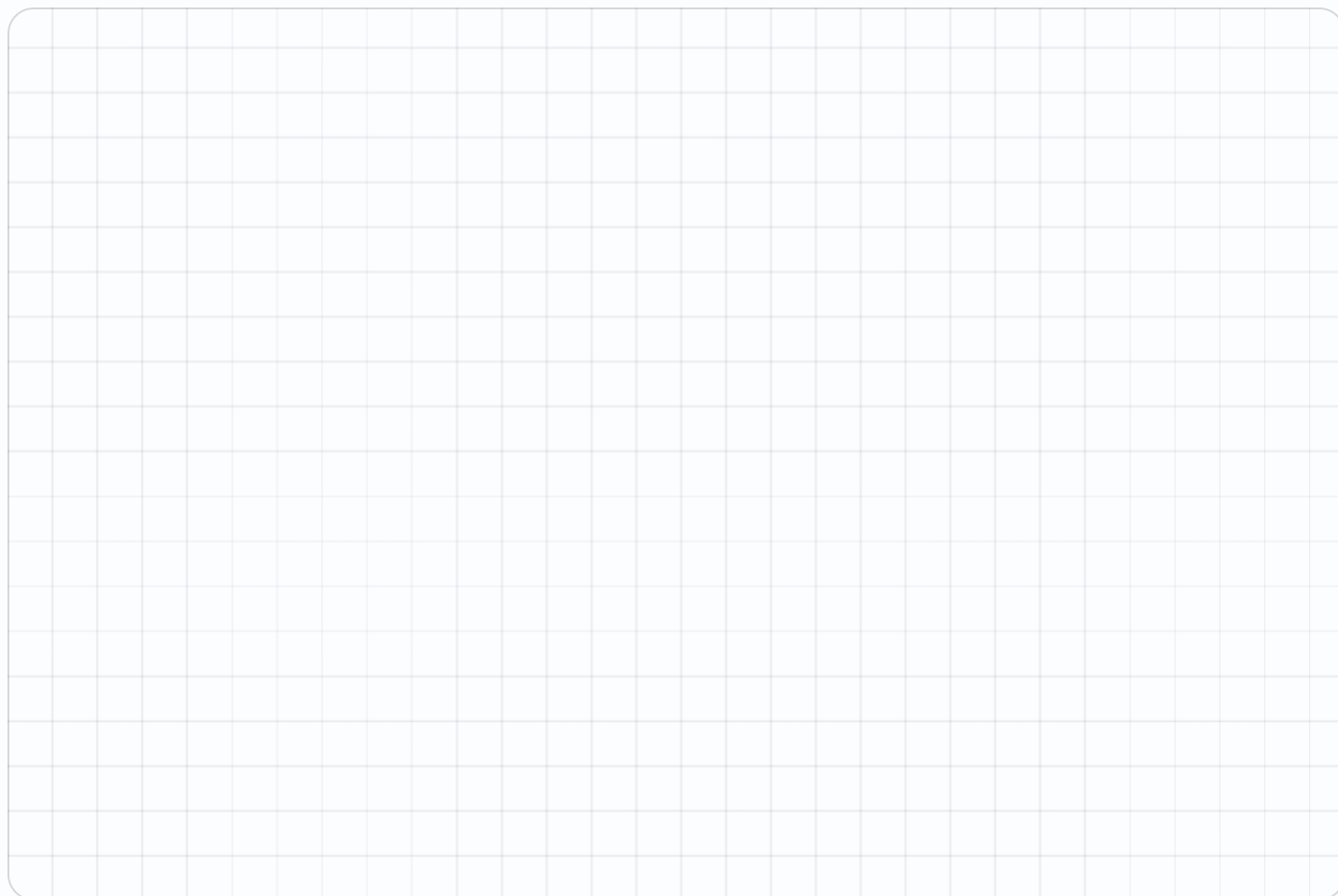
20. A $\frac{2}{3}$ by $\frac{3}{4}$ plot as a/b

★ Challenge NOT GRADED • REASON IT OUT

21. $\frac{2}{3} \times \frac{3}{4} \times \frac{1}{2}$
.....

22. A $\frac{2}{3}$ m by $\frac{3}{4}$ m bed — area
.....

23. $\frac{9}{16} \times \frac{8}{27}$
.....

24. How many $\frac{3}{8}$ m pieces from a 3 m ribbon
.....**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 5 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

5.1 • Finish the Garden Plan

OUT OF 18

Warm-Up • 1-12 • 10 | 6 | 9 | 8 | 12 | 6 | $\frac{1}{4}$ | 10 | 15 | 6 | 8 | 9

Core • 13-18 • 18 | 30 | $\frac{1}{2}$ | 24 | 35 | $\frac{1}{2}$

Challenge • 23-26 • $\frac{5}{9}$ | 16 | $\frac{1}{2}$ | 36

5.2 • Explain a Shrink

OUT OF 17

Warm-Up • 1-11 • 6 | 4 | 3 | 24 | 12 | 9 | 10 | 5 | half | 6 | 2

Core • 12-17 • $\frac{3}{4}$ | 12 | 75 | $\frac{5}{2}$ | 35 | smaller

Challenge • 22-25 • smaller | 50 | 1 | 63

5.3 • Multiply and Divide Together

OUT OF 16

Warm-Up • 1-10 • $\frac{1}{4}$ | 4 | 4 | $\frac{1}{6}$ | $\frac{1}{8}$ | 9 | 8 | 8 | $\frac{1}{8}$ | 6

Core • 11-16 • $\frac{1}{2}$ | 20 | 14 | $\frac{1}{12}$ | 20 | $\frac{1}{2}$

Challenge • 20-23 • $\frac{1}{4}$ | 12 | $\frac{3}{4}$ | 8

Thu • Error Journal Sweep

OUT OF 21

Warm-Up • 1-11 • $\frac{1}{8}$ | 4 | 8 | $\frac{1}{4}$ | 15 | $\frac{1}{10}$ | 5 | $\frac{1}{4}$ | 16 | 6 | 4

Core • 12-21 • $\frac{3}{10}$ | 21 | 20 | $\frac{1}{20}$ | 36 | $\frac{2}{5}$ | 28 | $\frac{1}{18}$ | 12 | 16

Challenge • 22-25 • $\frac{1}{6}$ | 24 | $\frac{1}{6}$ | $\frac{1}{15}$

Fri • Mission 06 Test

OUT OF 20

Warm-Up • 1-4 • $\frac{1}{6}$ | 5 | 7 | 6

Core • 5-20 • 8 | $\frac{1}{2}$ | $\frac{3}{10}$ | 16 | 25 | 20 | $\frac{1}{12}$ | 0.75 | $\frac{1}{3}$ | 18 | 27 | $\frac{1}{20}$
| $\frac{2}{3}$ | 24 | $\frac{5}{8}$ | $\frac{1}{2}$

Challenge • 21-24 • $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{6}$ | 8